

Prior Learning

Area of a parallelogram

$$A = bh$$

Area of a triangle

$$A = \frac{1}{2}bh$$

Area of a trapezoid

$$A = \frac{1}{2}(a + b)h$$

Area of a circle

$$A = \pi r^2$$

Circumference of a circle

$$C = 2\pi r$$

Volume of a cuboid

$$V = lwh$$

Volume of a cylinder

$$V = \pi r^2 h$$

Volume of a prism

$$V = Ah$$

Curved surface area of a cylinder

$$A = 2\pi rh$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint of a line segment

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Number and Algebra (SL & HL)

Arithmetic sequences

$$u_n = u_1 + (n - 1)d$$

$$S_n = \frac{n}{2}(2u_1 + (n - 1)d) = \frac{n}{2}(u_1 + u_n)$$

Geometric sequences

$$u_n = u_1 r^{n-1}$$

$$S_n = \frac{u_1(r^n - 1)}{r - 1} = \frac{u_1(1 - r^n)}{1 - r}, \quad r \neq 1$$

Compound interest

$$FV = PV \left(1 + \frac{r}{100k}\right)^{kn}$$

Exponents and logarithms

$$\log_a b = x \iff a^x = b, \quad a > 0, \quad b > 0, \quad a \neq 1$$

Laws of logarithms

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a x^m = m \log_a x$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Sum of an infinite geometric sequence

$$S_\infty = \frac{u_1}{1 - r}, \quad |r| < 1$$

Binomial theorem

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + b^n$$

$$\binom{n}{r} = \frac{n!}{r!(n - r)!}$$

Number and Algebra (HL only)

Combinations & permutations

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$${}^nP_r = \frac{n!}{(n-r)!}$$

Complex numbers

$$z = a + bi$$

Modulus–argument & exponential form

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta} = r \operatorname{cis} \theta$$

De Moivre's theorem

$$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta) = r^n e^{in\theta} = r^n \operatorname{cis} n\theta$$

Functions (SL & HL)

Equations of a straight line

$$y = mx + c$$

$$ax + by + d = 0$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Axis of symmetry of a quadratic

$$f(x) = ax^2 + bx + c \Rightarrow x = -\frac{b}{2a}$$

Solving a quadratic equation

$$ax^2 + bx + c = 0 \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta = b^2 - 4ac$$

Exponential and logarithmic functions

$$a^x = e^{x \ln a}$$

$$\log_a a^x = a^{\log_a x} = x, \quad a > 0, \quad a \neq 1$$

Functions (HL only)

Sum and product of roots of a polynomial

$$\sum_{r=0}^n a_r x^r = 0$$

$$\text{sum} = -\frac{a_{n-1}}{a_n}$$

$$\text{product} = (-1)^n \frac{a_0}{a_n}$$

Geometry and Trigonometry (SL & HL)

Distance & midpoint in 3D

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

Volumes & surface areas

$$V = \frac{1}{3}Ah \quad (\text{pyramid})$$

$$V = \frac{1}{3}\pi r^2 h \quad (\text{cone})$$

$$A = \pi r l \quad (\text{cone, curved surface})$$

$$V = \frac{4}{3}\pi r^3 \quad (\text{sphere})$$

$$A = 4\pi r^2 \quad (\text{sphere, surface})$$

Sine rule, cosine rule, area of a triangle

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$A = \frac{1}{2}ab \sin C$$

Arc length & sector area

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

Tangent identity

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Pythagorean & double angle identities

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

Geometry and Trigonometry (HL only)

Reciprocal & Pythagorean identities

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Compound & double angle identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Magnitude of a vector

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Scalar product & angle between vectors

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + v_3 w_3 = |\mathbf{v}| |\mathbf{w}| \cos \theta$$

$$\cos \theta = \frac{v_1 w_1 + v_2 w_2 + v_3 w_3}{|\mathbf{v}||\mathbf{w}|}$$

Vector, parametric & Cartesian line equations

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$$

$$x = x_0 + l\lambda, \quad y = y_0 + m\lambda, \quad z = z_0 + n\lambda$$

$$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$$

Vector product & area of a parallelogram

$$\mathbf{v} \times \mathbf{w} = (v_2 w_3 - v_3 w_2, \quad v_3 w_1 - v_1 w_3, \quad v_1 w_2 - v_2 w_1)$$

$$|\mathbf{v} \times \mathbf{w}| = |\mathbf{v}||\mathbf{w}| \sin \theta$$

$$A = |\mathbf{v} \times \mathbf{w}|$$

Vector & Cartesian equations of a plane

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b} + \mu \mathbf{c}$$

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$$

$$ax + by + cz = d$$

Statistics and Probability (SL & HL)

Interquartile range

$$IQR = Q_3 - Q_1$$

Mean of a set of data

$$\bar{x} = \frac{\sum f_i x_i}{n}, \quad n = \sum f_i$$

Probability & complementary events

$$P(A) = \frac{n(A)}{n(U)}$$

$$P(A) + P(A') = 1$$

Combined, conditional & independent events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) \quad (\text{mutually exclusive})$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(A)P(B) \quad (\text{independent})$$

Expected value of a discrete random variable

$$E(X) = \sum xP(X = x)$$

Binomial distribution

$$X \sim B(n, p)$$

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

Standardized normal variable

$$z = \frac{x - \mu}{\sigma}$$

Statistics and Probability (HL only)

Bayes' theorem

$$P(B | A) = \frac{P(A | B) P(B)}{P(A | B) P(B) + P(A | B') P(B')}$$

$$P(B_i | A) = \frac{P(B_i)P(A | B_i)}{\sum_j P(B_j)P(A | B_j)}$$

Variance & standard deviation

$$\sigma^2 = \frac{\sum f_i(x_i - \mu)^2}{n} = \frac{\sum f_i x_i^2}{n} - \mu^2$$

$$\sigma = \sqrt{\frac{\sum f_i(x_i - \mu)^2}{n}}$$

Linear transformation of a random variable

$$E(aX + b) = aE(X) + b$$

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Continuous random variable X — expectation and variance

$$E(X) = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Variance of discrete & continuous X

$$\text{Var}(X) = \sum (x - \mu)^2 P(X = x) \quad (\text{discrete})$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx \quad (\text{continuous})$$

Calculus (SL & HL)

Derivative of x^n

$$f(x) = x^n \Rightarrow f'(x) = nx^{n-1}$$

Integral of x^n & area under a curve

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$A = \int_a^b y dx, \quad f(x) > 0$$

Standard derivatives

$$f(x) = \sin x \Rightarrow f'(x) = \cos x$$

$$f(x) = \cos x \Rightarrow f'(x) = -\sin x$$

$$f(x) = e^x \Rightarrow f'(x) = e^x$$

$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$

Chain, product & quotient rules

$$y = g(u) \Rightarrow \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad (\text{chain})$$

$$y = uv \Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} \quad (\text{product})$$

$$y = \frac{u}{v} \Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad (\text{quotient})$$

Kinematics

$$a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$$

$$\text{distance} = \int_{t_1}^{t_2} |v(t)| dt$$

$$\text{displacement} = \int_{t_1}^{t_2} v(t) dt$$

Standard integrals

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int e^x dx = e^x + C$$

Area enclosed by a curve and the x -axis

$$A = \int_a^b y dx$$

Calculus (HL only)

Derivative from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Further standard derivatives

$$\tan x \rightarrow \sec^2 x$$

$$\sec x \rightarrow \sec x \tan x$$

$$\operatorname{cosec} x \rightarrow -\operatorname{cosec} x \cot x$$

$$\cot x \rightarrow -\operatorname{cosec}^2 x$$

$$a^x \rightarrow a^x \ln a$$

$$\log_a x \rightarrow \frac{1}{x \ln a}$$

$$\arcsin x \rightarrow \frac{1}{\sqrt{1-x^2}}$$

$$\arccos x \rightarrow \frac{-1}{\sqrt{1-x^2}}$$

$$\arctan x \rightarrow \frac{1}{1+x^2}$$

Further standard integrals

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan \frac{x}{a} + C$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin \frac{x}{a} + C, \quad |x| < a$$

Integration by parts

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int u dv = uv - \int v du$$

Area & volume of revolution

$$A = \int_a^b x dy \quad (\text{area against the } y\text{-axis})$$

$$V = \pi \int_a^b y^2 dx \quad (\text{revolution about } x\text{-axis})$$

$$V = \pi \int_a^b x^2 dy \quad (\text{revolution about } y\text{-axis})$$

Euler's method & integrating factor

$$y_{n+1} = y_n + h f(x_n, y_n), \quad x_{n+1} = x_n + h$$

$$\text{integrating factor} = e^{\int P(x) dx} \quad \text{for } y' + P(x)y = Q(x)$$

Maclaurin series

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$